



VISHMI BHAGYA EDIRISINGHE

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PROFESSIONAL SUMMARY

Detail-oriented Data Science undergraduate at SLIIT (GPA: 3.8/4.0) with strong analytical, technical, and reporting skills. Experienced in data analysis, dashboard creation, AI-driven systems, and full-stack application development. Proficient in Python, SQL, Excel, and data visualization tools, with hands-on exposure to machine learning, NLP, and agent-based systems. Active researcher with publications and commissioned studies conducted for the Sri Lanka Police, and an abstract presented at an international conference on child protection.

EDUCATION

BSc. (Hons) in Information Technology — Data Science | *Sri Lanka Institute of Information Technology (SLIIT)* 2023 – Present

- Currently in 3rd Year | Expected Graduation: 2027
- Cumulative GPA: 3.8 / 4.0
- Key coursework: Machine Learning, Data Mining, Database Systems, Statistics, Software Engineering

G.C.E. Advanced Level (A/L) | *R/Sumana Balika Vidyalaya, Ratnapura* 2022

G.C.E. Ordinary Level (O/L) | *R/Sumana Balika Vidyalaya, Ratnapura* 2019

RESEARCH EXPERIENCE

Co-Author & Research Assistant | *Sri Lanka Police (Commissioned Research)* 2024 – 2025

- Contributed as co-author and research assistant on two studies commissioned by the Sri Lanka Police
- Participated in all phases of the research process: study design, primary data collection, field surveys, data analysis, and report writing
- Research 1: An Analytical Study of Statutory Rape Victims in Sri Lanka — examined victim profiles, patterns, and contributing factors using national case data; findings submitted to the Sri Lanka Police
- Research 2: A Community-Based Survey on Increasing Educational Qualification Requirements for Police Recruitment in Sri Lanka — assessed community perspectives on raising entry standards for police personnel

Data Collector — SGBV Research Project | *Faculty of Medicine, UoK* 2025 – 2026

- Participating in a data collection and data analysis initiative on Sexual and Gender-Based Violence (SGBV) coordinated by the Faculty of Medicine, UoK
- Collecting primary data from Sri Lanka Police records in support of the research team's fieldwork
- Handling sensitive data with confidentiality and adherence to ethical research standards

PUBLICATIONS & PRESENTATIONS

[1] Youth at Risk: Exploring Causes and Methods of Child Suicide Through National Data in Sri Lanka

Author: E.A.V.B. Edirisinghe | **Co-Author:** E.A.D.A. Edirisinghe

Presented at: *International Conference on Child Protection 2025 (ICCP 2025)*

Venue: Faculty of Computing & Technology, University of Kelaniya, Sri Lanka | 2025

[2] An Analytical Study of Statutory Rape Victims in Sri Lanka

Role: Co-Author & Research Assistant

Commissioned by: Sri Lanka Police | Submitted to the Sri Lanka Police (Not publicly published)

[3] A Community-Based Survey on Increasing Educational Qualification Requirements for Police Recruitment in Sri Lanka

Role: Co-Author & Research Assistant

Commissioned by: Sri Lanka Police | Submitted to the Sri Lanka Police (Not publicly published)

TECHNICAL SKILLS

Data Analysis & Reporting: Microsoft Excel (Pivot Tables, Dashboards, Data Validation), Google Sheets, Exploratory Data Analysis (EDA), Data Cleaning & Preprocessing, KPI Tracking, Statistical Analysis

Programming Languages: Python (Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn), SQL (Relational, ORDB, Advanced Queries, Schema Design), R (Statistical Analysis)

Databases & Data Stores: MySQL, MongoDB, SQL Server (MSSQL)

Data Engineering & Warehousing: ETL/ELT Pipeline Development, Data Warehousing (Star Schema, Fact & Dimension Tables, OLAP vs OLTP), Data Modelling, Database Design & Normalization (1NF, 2NF, 3NF), ER Modelling, Data Pipeline Architecture, Data Integration, Business Intelligence Reporting

AI & Machine Learning: Supervised & Unsupervised Learning, Feature Engineering, Model Evaluation & Validation (Precision, Recall, F1, AUC-ROC), NLP Fundamentals, Sentence-BERT & Embeddings, Prompt Engineering, Agent-Based Systems

Visualization & BI Tools: Power BI (Dashboard Development, Data Visualization, KPI Reporting), Matplotlib, Seaborn, Streamlit

Development & Frameworks: Flask, MERN Stack (MongoDB, Express, React, Node.js), SQLite, Spring Boot, REST APIs

Developer Tools: Google Colab, VS Code, IntelliJ IDEA, GitHub, Git, SQL Server Management Studio

Cloud Platforms: AWS (S3, Databases, Auto-Scaling, Networking), Microsoft Azure (Virtual Machines, Storage, Azure Data Services, Azure Data Factory), Microsoft Fabric (Exposure)

Professional Skills: Technical Documentation, Report Writing, Research & Data Collection, Strong Written & Verbal Communication (English), Attention to Detail, Team Collaboration, Fast Learner

PROJECTS

Coffee Shop Data Warehouse & Business Intelligence Solution

March 2026

- Designed an end-to-end Data Warehouse using **Kimball Star Schema** with a central fact table, five dimension tables, and **SCD Type 2** for customer history tracking
- Built multi-source **ETL pipelines in SSIS** extracting from SQL Server, CSV, and TXT sources through a staging layer, applying Merge Join, Lookup, Stored Procedures, and Derived Column transformations
- Built an **SSAS OLAP Cube** with multi-level hierarchies supporting Roll-up, Drill-down, Slice, Dice, and Pivot via Excel PivotTables
- Developed and published **four Power BI reports** to Power BI Service with DAX measures, Matrix views, Cascading Slicers, and Drill-through capabilities

Technologies: SQL Server, SSIS, SSAS, Power BI, Excel, T-SQL, DAX, SSMS

Power BI Dashboard | Fitness & Performance Tracking

December 2025

[View Power BI Dashboard – Fitness Performance](#)

- Built an interactive Power BI dashboard to analyze workout patterns, activity levels, and performance trends
- Designed KPI-based visualizations to monitor metrics such as calories burned, duration, and workout consistency
- Enabled dynamic filtering and drill-down capabilities for personalized data exploration
- Transformed raw fitness data into actionable insights to improve performance and goal tracking

Technologies: Power BI

Household Finance Management System | AI-Powered Reporting Platform (Group)

Aug – Oct 2025
github.com/AMODHYAJ/household_finance_app-final

- Built a multi-agent AI system for automated financial data analysis and reporting
- Implemented data validation, transaction categorization, and anomaly detection modules
- Designed interactive dashboards for spending trend visualization and financial summaries
- Generated intelligent financial insights and actionable recommendations using AI agents

Technologies: Python, Flask, SQLite, Data Visualization

SecureBank AI — Credit Card Fraud Detection | *ML Pipeline (Group)* Aug – Oct 2025

github.com/AMODHYAJ/Y3S1-Credit-Card-Fraud-Detection

- Developed an end-to-end machine learning pipeline for real-time credit card fraud detection
- Performed feature engineering and applied techniques to handle class imbalance (SMOTE)
- Trained and compared Isolation Forest, Random Forest, and Logistic Regression models
- Evaluated model performance using precision, recall, F1-score, and AUC-ROC metrics

Technologies: Python, Scikit-learn, Pandas, NumPy, Matplotlib

PDN Products — Furniture Manufacturing & Retail System | *Full-Stack Project* Feb – May 2025

github.com/AMODHYAJ/PDN_Furniture

- Designed and developed product, inventory, and order management modules with the MERN stack
- Performed data validation, multi-record entry, and consistency checks across structured datasets
- Generated categorized performance reports for order flow and product analytics

Technologies: MongoDB, Express.js, React, Node.js (MERN Stack)

Police Education Data Analysis | *Public Dataset Analysis*

September 2025

github.com/vishmi11/police-education-analysis

- Cleaned and preprocessed raw public datasets for analysis of police education and training records
- Conducted EDA to identify patterns in training participation and performance outcomes
- Produced visual reports and charts to support data-driven decision-making

Technologies: Python, Pandas, Matplotlib

Equipment Maintenance & Performance Analysis | *Operational Data Analysis* December 2025

github.com/vishmi11/Equipment_Maintenance_Analysis

- Conducted data cleaning and exploratory analysis on machine maintenance and production logs
- Calculated Overall Equipment Effectiveness (OEE) metrics to assess and compare machine performance
- Identified downtime patterns, trends, and optimization opportunities using statistical analysis
- Developed interactive dashboards using Excel to visualize equipment performance, downtime, and OEE metrics

Technologies: Python, Pandas, Matplotlib, Seaborn, Excel

CERTIFICATIONS

- AWS SimuLearn Fundamentals — Amazon Web Services (Cloud, Security, Networking, Databases, S3)
- Microsoft Azure Fundamentals — Virtual Machines, Networking, Storage, Security
- Prepare Data for Analysis with Power BI — Microsoft Learn
- Mongo DB Python Developer Path
- Introduction to Generative AI — Google Cloud Skills Boost
- English Proficiency: Upper Secondary Plus B1.1 — British Council

ACHIEVEMENTS

- **Dean's List — Faculty of Computing, SLIIT** | Year 1 Semester 1 (3.76), Year 1 Semester 2 (3.75), Year 2 Semester 1 (3.88), Year 2 Semester 2 (3.86)

VOLUNTEER EXPERIENCE & LEADERSHIP

Member | *SLIIT FOSS Women Community*

2025 – Present

- Actively participated in women-in-tech community initiatives
- Served as an Organizing Committee Member for ImpactX, contributing to event planning and execution

Executive Member | *Youth Committee — Community Crime Prevention Foundation* 2025 - Present

- Contributed to community safety programs and youth engagement activities

Volunteer | *Centre for Gender Studies, University of Kelaniya*

2023 - Present

- Supported gender equality research and awareness programs
- Volunteered at the International Conference on Gender Disparities, assisting with conference coordination and operations

Senior Prefect | *R/Sumana Balika Vidyalaya*

2021 – 2022

- Demonstrated leadership, discipline, and responsibility as a senior school prefect
- Junior Prefect: 2018 – 2019